

This manuscript investigates the endurance degradation of HZO/SiO₂/Si FeFETs by comparing the PUND-extracted polarization with the DC C–V hysteresis window. The observation that ΔV_{hyst} approaches zero and subsequently changes sign while 2Pr remains higher than its pristine value is interesting and suggests that polarization loss alone cannot account for the early degradation. However, the present results do not directly demonstrate the loss of the transistor memory window or independently quantify the proposed trapped charge. In addition, the C–V sign reversal is observed under a high-loss condition where the adopted equivalent-circuit correction is no longer valid. The following major issues should be addressed before the manuscript can be considered for publication.

1. The FeFET memory-window failure should be verified using transistor readout measurements. The endurance conclusion is currently based on slow-sweep DC C–V hysteresis. However, the DC C–V window is sensitive to sweep rate, dwell time, minority-carrier response, and charge trapping/detrapping, and is not necessarily equivalent to the program/erase threshold-voltage window. The authors should provide I_D - V_G , V_{th} , or fixed- V_G read-current data after program and erase operations as a function of cycle number, with the read delay and extraction method clearly specified. If such data are unavailable, the claims should be limited to the degradation of the DC C–V hysteresis window rather than FeFET memory failure or loss of stored information.
2. The interpretation of the charge density estimated using Eq. (9) should be clarified.

The reported $\Delta N_{\text{it}} \approx 1.3 \times 10^{13} \text{ cm}^{-2}$ is estimated from the measured window shift using $C_{\text{IL}} \Delta V / q$ and therefore represents an equivalent charge-density estimate rather than a direct extraction of interface-trap density. The authors should clarify the assumptions of this conversion and consider using a notation such as ΔN_{eq} . To support the proposed trapping mechanism, at least one independent trap-sensitive measurement should be provided. A practical approach would be to measure the recovery of V_{th} or I_D as a function of read delay after program and erase operations at selected cycling checkpoints, such as the pristine state, 10^3 and 10^4 cycles. Charge-pumping measurements may be used as an alternative if available. If no independent trap-sensitive data can be provided, the mechanism should be presented as being consistent with charge trapping rather than as a direct identification of the responsible traps.

3. The roles of wake-up, charge trapping, and fatigue should be discussed more carefully. The increase in 2Pr up to approximately 10^2 cycles is consistent with the commonly reported wake-up behavior of HZO. Thus, polarization wake-up and increasing charge trapping may occur simultaneously during the early cycling stage. At 10^5 cycles, the reduction in 2Pr, broadening of the switching peaks, and increase in non-switching current indicate that ferroelectric degradation and electrical damage may also contribute at later stages. Therefore, the statement that endurance failure is “not” caused by ferroelectric fatigue is too general. The data more specifically show that loss of switchable polarization alone cannot explain the early collapse of the DC C–V window in the investigated device. The Introduction and Abstract should also better acknowledge previous reports identifying charge trapping, trap generation, and FE/DE interfacial-layer effects as important FeFET endurance mechanisms.
4. The limitations regarding charge location, response time, and reproducibility should be more clearly stated.

The present measurements do not uniquely distinguish charge located at the HZO/SiO₂ interface, within the SiO₂ interlayer, at the SiO₂/Si interface, or inside HZO. Unless additional localization

evidence is available, a more general description, such as “effective charge within the channel-side gate stack,” would be more appropriate. Moreover, the 4.7-6.0 μ s time constant obtained from the AC dispersion is substantially shorter than the approximately 17 ms dwell time of the DC C–V measurement. The manuscript should therefore avoid directly assigning these two responses to the same charge population unless further time-dependent evidence is provided. Finally, the authors should clarify whether the window collapse and sign reversal were reproduced in additional devices or independent endurance runs. If the results are limited to a single device, this limitation should be explicitly stated and the generality of the conclusions should be moderated.